

Homework 4

Deadline: March 17, 2022

Exercise 4.6a

4.6 (a) A sample consisting of 1.00 mol of a monatomic perfect gas with $C_{V,m} = \frac{3}{2}R$ is heated from 100°C to 300°C at constant pressure. Calculate ΔS (for the system).

Exercise 4.7a

4.7 (a) Calculate ΔS (for the system) when the state of 3.00 mol of a monatomic perfect gas, for which $C_{p,m} = \frac{5}{2}R$, is changed from 25°C and 1.00 atm to 125°C and 5.00 atm. How do you rationalize the sign of ΔS ?

Exercise 4.8a

4.8 (a) A sample consisting of 3.00 mol of a diatomic perfect gas at 200 K is compressed reversibly and adiabatically until its temperature reaches 250 K. Given that $C_{V,m} = 27.5 \text{ J K}^{-1} \text{ mol}^{-1}$, calculate q , w , ΔU , ΔH , and ΔS .

Exercise 4.11b

4.11 (b) A sample of copper of mass 2.75 kg is cooled at constant pressure from 330 K to 275 K. Calculate (a) the energy that must be removed as heat and (b) the change in entropy of the sample.

Exercise 4.16a

4.16 (a) Consider a system consisting of 2.0 mol $\text{CO}_2(\text{g})$, initially at 25°C and 10 atm and confined to a cylinder of cross-section 10.0 cm^2 . It is allowed to expand adiabatically against an external pressure of 1.0 atm until the piston has moved outwards through 20 cm. Assume that carbon dioxide may be considered a perfect gas with $C_{V,m} = 28.8 \text{ J K}^{-1} \text{ mol}^{-1}$ and calculate (a) q , (b) w , (c) ΔU , (d) ΔT , (e) ΔS .

Exercise 4.23a

4.23 (a) Calculate the change in the entropies of the system and the surroundings, and the total change in entropy, when a sample of nitrogen gas of mass 14 g at 298 K and 1.00 bar doubles its volume in (a) an isothermal reversible expansion, (b) an isothermal irreversible expansion against $p_{\text{ex}} = 0$, and (c) an adiabatic reversible expansion.